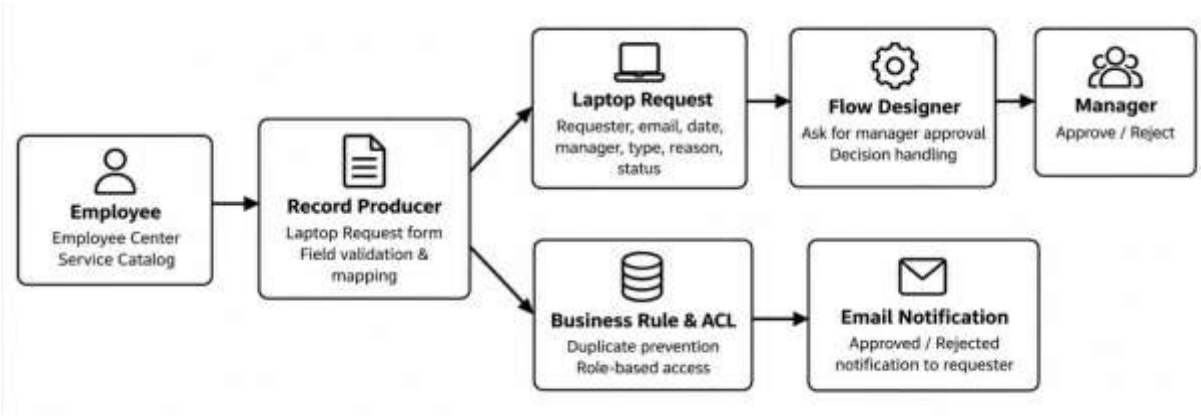
Project Design Phase-II
Data Flow Diagram & User Stories

Date	8 September 2026
Team ID	SWUID2026190365
Project Name	ServiceNow-Based Employee Laptop Request Application
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) represents how information enters, moves through, is processed, and is stored in the system. For this project, the DFD shows an employee submitting a laptop request through ServiceNow, the request being stored, the manager approval workflow being triggered, and the requester receiving the final status notification.



Project DFD: Employee → ServiceNow Request → Laptop Request Table → Manager Approval → Status & Email.

User Stories

The following user stories describe the main functions of the ServiceNow-based laptop request application.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Employee	Laptop Request	USN-1	As an employee, I can submit a laptop request through the ServiceNow Employee Center.	A laptop request record is created with the required request details.	High	Sprint-1
Employee	Laptop Request	USN-2	As an employee, I can select the type of laptop required (Basic or Advance).	The selected laptop type is stored with the request.	High	Sprint-1

Employee	Request Tracking	USN-3	As an employee, I can view the status of my laptop request.	The request status shows Pending, Approval, Approved, or Rejected.	High	Sprint-1
Manager	Approval	USN-4	As a manager, I can review and approve or reject an employee laptop request.	The approval record is assigned to the manager and the decision is recorded.	High	Sprint-1
System	Duplicate Prevention	USN-5	As the system, I prevent the same employee from submitting more than one laptop request in a year.	A duplicate request is blocked with an appropriate message.	High	Sprint-1
Employee	Notification	USN-6	As an employee, I receive an email when my laptop request is approved or rejected.	The requester receives the decision notification by email.	Medium	Sprint-2
IT Manager	Monitoring	USN-7	As an IT manager, I can view laptop requests for tracking and accountability.	Authorized managers can read the laptop request records.	Medium	Sprint-2

SERVICENOW

Employee Laptop Request

Request, Approval, Fulfillment and Support for a New Laptop

Empowering Employees with the Right Technology for a Better Tomorrow



SCENARIO

Requesting, approving, receiving and using a laptop through ServiceNow

	Entice	Enter	Engage	Exit	Extend
	How does someone initially become aware of this process?	What do people experience as they begin the process?	In the core moments in the process, what happens?	What do people typically experience as the process finishes?	What happens after the experience is over?
Steps What does the person (or group) typically experience?	Learn about laptop request process	Access ServiceNow portal	Log in to ServiceNow	Open Laptop Request form	Fill in required details	Submit request	Manager review & approval	IT team processing	Laptop procurement	Receive laptop	Acknowledge receipt	Use laptop for work	Raise support (if needed)	Future upgrades/ replacement
Interactions What interactions do they have at each step along the way? • **People:** Who do they see or talk to? • **Places:** Where are they? • **Things:** What digital touchpoints or physical objects would they use?	Emails, announcements, company policies	ServiceNow Employee Portal	Employee credentials (Single Sign-On)	ServiceNow Catalog	Laptop request form (hardware catalog item)	Submit button & confirmation message	Manager (approval task)	IT team (ServiceNow workflow)	Vendor / Procurement system	E-mail notification	ServiceNow portal (request status)	Physical laptop	IT support (ServiceNow case)	ServiceNow (upgrade / replacement request)
Goals & motivations At each step, what is a person's primary goal or motivation? ("Help me..." or "Help me avoid...")	Help me understand how to request a laptop	Help me access the right portal quickly	Help me log in without issues	Help me find the correct request form	Help me provide accurate information	Help me submit my request easily	Help me get my request approved	Help me ensure my laptop is processed quickly	Help me receive the right laptop on time	Help me know when my laptop is delivered	Help me confirm successful delivery	Help me use my laptop productively	Help me get support if I face any issues	Help me request an upgrade or replacement in future
Positive moments What steps does a typical person find enjoyable, productive, fun, motivating, delightful, or exciting?	Clear communication about the process	Easy access to ServiceNow	Quick and seamless login		Simple and user-friendly request form	Instant submission confirmation	Quick approval by manager	Timely processing by IT team	Receiving the laptop on time	Clear notifications at each stage	Successfully receiving and setting up the laptop	Productive work with new laptop		Option to request upgrades easily
Negative moments What steps does a typical person find frustrating, confusing, angering, costly, or time-consuming?	Lack of clarity about the process	Difficulty accessing the portal	Login issues (SSO errors)	Confusion about which form to select	Too many mandatory fields		Delays in manager approval	Long processing time by IT	Stock unavailability or wrong laptop delivered	No status updates	Issues with laptop setup	Delay in support response		Uncertainty about upgrade eligibility
Areas of opportunity How might we make each step better? What ideas do we have? What have others suggested?	Create awareness campaigns and guidelines	Provide a direct link on company intranet	Enable smooth SSO login experience	Simplify the catalog and improve search	Auto-fill user details where possible	Provide instant confirmation and tracking	Optimize approval workflow (with SLAs)	Integrate with inventory for real-time stock updates	Send proactive notifications (email/portal)		Provide setup guides and IT support resources	Offer self-service troubleshooting		Allow easy upgrade/ replacement requests

Solution Requirements (Functional & Non-functional)

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Functional Requirements:

Following are the functional requirements of the proposed ServiceNow laptop request solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Laptop Request Submission	Employee submits a laptop request using the ServiceNow Employee Center / Service Catalog Record Producer.
FR-2	Request Details	Capture requester, employee email, request date, short description, manager, laptop type, reason and status.
FR-3	Duplicate Prevention	Allow only one laptop request per employee per calendar year.
FR-4	Manager Approval	Automatically send the request to the selected manager through Flow Designer for approval or rejection.
FR-5	Status Management	Update the request status to Pending, Approved or Rejected according to the workflow outcome.
FR-6	Notifications	Send an email notification to the requester after approval or rejection.
FR-7	Access Control	Allow employees to create/read/write requests and managers to view requests according to assigned roles and ACLs.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The request form should be simple, clear and easy for employees to complete.
NFR-2	Security	Role-based access and ACLs restrict who can create, read and modify laptop request

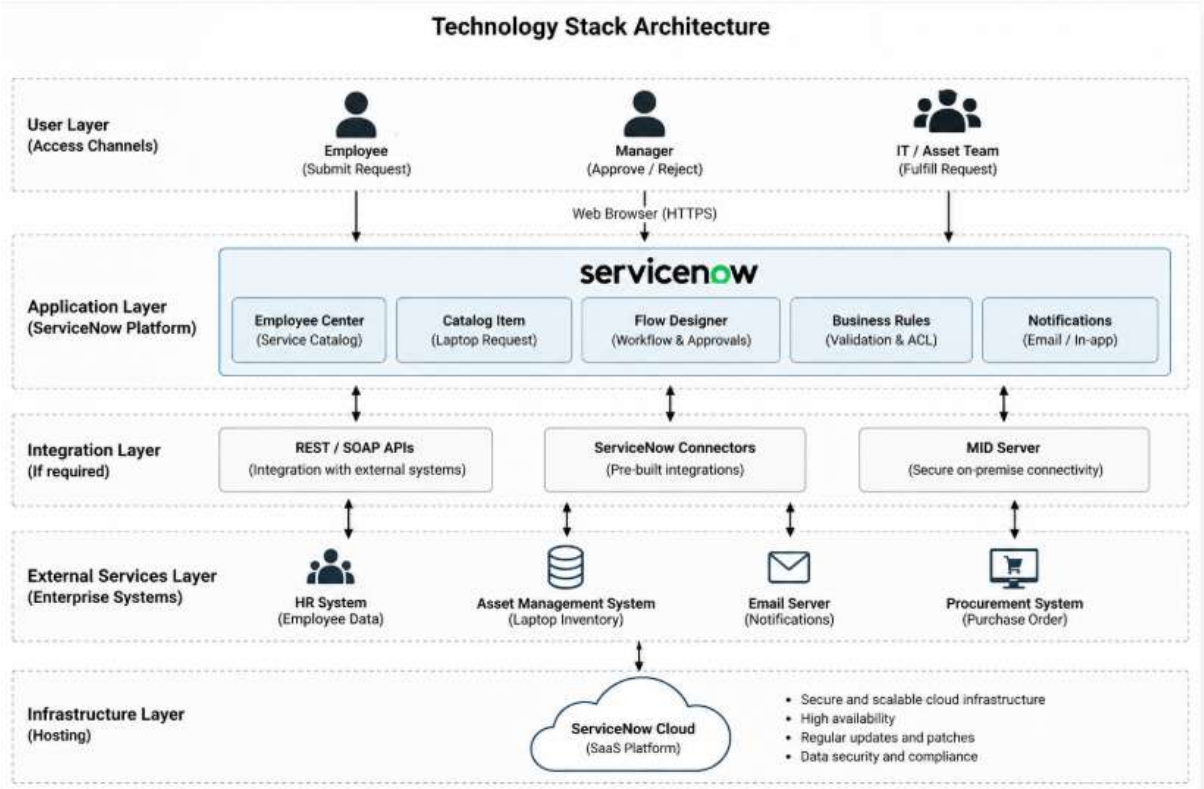
		records.
NFR-3	Reliability	The approval flow and status updates should execute consistently without manual intervention.
NFR-4	Performance	The form, request lookup and approval workflow should respond within normal ServiceNow application performance expectations.
NFR-5	Availability	The application should be available through the organization's ServiceNow instance during supported service hours.
NFR-6	Scalability	The centralized table and Flow Designer workflow should support increasing numbers of employees and requests.

Technology Stack (Architecture & Stack)

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Technical Architecture:

The architecture uses ServiceNow as the centralized platform for employee laptop requests. Employees submit requests through the Employee Center/Service Catalog. The Record Producer creates a record in the custom Laptop Request Form table. Flow Designer sends the request to the manager, waits for approval or rejection, updates the status, and sends an email notification.



Project Technical Architecture

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Employee and manager interaction with the application.	ServiceNow Employee Center / Service Catalog
2	Application Logic-1	Creates and manages laptop request records.	ServiceNow Custom Table & Record Producer
3	Application Logic-2	Automates manager approval and decision handling.	ServiceNow Flow Designer
4	Application Logic-3	Prevents duplicate yearly requests.	ServiceNow Business Rule
5	Database	Stores laptop request	ServiceNow platform

		and related application data.	database / custom table
6	Cloud Database	Not separately required; data is maintained in the ServiceNow platform.	ServiceNow data platform
7	File Storage	No separate file-storage component is required for the core application.	ServiceNow platform
8	External API-1	Not required for the core laptop request workflow.	Not applicable
9	External API-2	Not required for the core laptop request workflow.	Not applicable
10	Machine Learning Model	No machine-learning model is required.	Not applicable
11	Infrastructure (Server / Cloud)	Application is hosted and executed in the organization's ServiceNow instance.	ServiceNow cloud platform

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No external open-source application framework is required for the core ServiceNow configuration.	ServiceNow platform configuration
2	Security Implementations	Role-based access is implemented using it_manager and it_employee roles, groups and ACLs.	ServiceNow Roles, Groups and ACLs
3	Scalable Architecture	Centralized table and workflow-based architecture can support additional employees and requests.	ServiceNow platform & Flow Designer
4	Availability	The application is delivered through the organization's ServiceNow instance.	ServiceNow cloud platform
5	Performance	Use simple forms, indexed platform fields and automated Flow Designer processing; avoid unnecessary manual	ServiceNow platform

		steps.	

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